

A spatial–temporal model for network-wide flight delay prediction based on federated learning

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ABSTRACT

Network-wide flight delay prediction has emerged as a severe and prominent issue within the realm of intelligent aviation systems, owing to its crucial impact on flight scheduling, airline planning, and airport operations. The methodology employed in flight delay prediction has undergone substantial evolution in recent years, progressing from rudimentary statistical models to more sophisticated and intricate deep learning models. Nevertheless, the majority of current research endeavors primarily concentrate on the utilization of all available local data to collectively construct a prediction model, enduring significant challenges pertaining to data security and privacy issues as well as enormous communication overhead. To tackle these challenges, this investigation centers its attention on developing a hybrid federated deep learning model (HFDL), which only emphasizes updating the model parameters exclusively on the central server without the need for accessing or sharing any private data. Firstly, we employ the Louvain algorithm to partition the global spatial graph of the airport network into several sub-regions. Then, we present a local training model for each sub-region, which uses a diffusion graph convolutional network (DGCN) and residual gated recurrent unit (RGRU) to model the complex local temporal and spatial dependencies of the airport network. The model utilizes a branch structure to predict the flight delay at each airport within a sub-region. Finally, the local models are aggregated through federated learning to form a vigorous central model that bridges the limits on global data sharing and privacy guarantee. Experiments are carried out on a real flight delay dataset from the U.S. Bureau of Transportation Statistics (BTS) database with the aim of analyzing the performance of the proposed prediction model. The experimental results demonstrate that the HFDL model can attain excellent prediction performance without the interaction of sub-region raw data and improve training efficiency. Moreover, the model has strong generalization ability. The visualization results demonstrate that the proposed model can effectively explore the spatial–temporal dependencies of flight delays, providing guidance for formulating air traffic control strategies.

1. Introduction

With the ever-increasing demand for air transportation, modern aviation systems are under considerable strain, and limited flight space resources are further compressed, resulting in widespread and frequent occurrences of network-wide flight delays. Flight delay is looked upon as a chief performance indicator of aviation systems. As per the regulation of the Civil Aviation Administration of China (CAAC), flights arriving over 15 min later than scheduled time are viewed as delayed flights. Following this rule and the statistics, the delay rates of major Chinese airlines (e.g., Air China, China Southern, and China Eastern) were between 27% and 36% in 2016, whereas their U.S. counterparts

(e.g., United, Delta, American, and Southwest) were between 13% and 20% [1,2]. The high flight delays pose inevitable consequences and plague airlines and passengers chronically, such as economic loss as well as dissatisfied passenger experiences [3,4]. It is estimated that in 2019, the cost caused by flight delays to the global economy was around \$50 billion [5]. Such high economic losses and dissatisfied passenger experiences motivate the research community to conduct relevant research on flight delay prediction. Accurate flight delay prediction cannot only assist aviation authorities and airport regulators in devising more efficient air traffic control strategies to alleviate flight delays and increase airport productivity but also help informed

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travelers adjust their pre-trip plans and mitigate their unnecessary anxiety.

Flight delay prediction utilizes historical flight data to build a prediction model to estimate and predict future flight delays. Conventional flight delay prediction methods typically rely on mathematical simulation models, e.g., periodic functions [6], polynomial spline [7], two-stage least squares [8], etc. These models have the advantages of being simple, fast, and requiring just a few pieces of data. Nonetheless, these models have poor expressive capacity and weak fitting ability. Nowadays, the advanced Aircraft Condition Monitoring System (ACMS) is connected to a large number of sensors and detectors distributed on the aircraft to collect more and more flight data. With the increase in data storage and computation capabilities, data-driven machine learning approaches for flight delay prediction have shown more promise than conventional approaches. Since flight delay prediction subsumes two main research streams, i.e., single-object prediction and network-wide prediction, and different research streams have distinct data-driven machine learning approaches. In single-object prediction, it encompasses the study of single-airport-level prediction using methods such as naive Bayes, deep belief network [9,10], and single-airline-level prediction using random forest, K-nearest neighbor, gradient boosting classifier, and supervised neural network [11,12]. Different from single-object prediction, network-wide flight delay prediction exhibits a complex spatial dependency caused by the interconnectedness of airports across various regions. For example, the occurrence of the flight delay at one airport has the potential to result in delays at both upstream and downstream airports within the entire airport network [13]. In other words, how to explore the spatial-temporal correlations at the network level is vital for forecasting flight delay. As the airport network has graph-structured data, graph neural network-based approaches have emerged as a highly effective means of extracting temporal and spatial patterns inherent in airport flight delay. Several distinct graph-based approaches, such as graph convolutional network (GCN) [14,15] and graph attention network (GAT) [16,17], have been developed and utilized to effectively capture extensive temporal and spatial patterns in the context of flight delay prediction.

Nevertheless, the aforementioned deep learning models are performed in a centralized fashion, which necessitates uploading flight delay data information collected by diverse sensors to the cloud. Despite the fact that it is possible to train a central model with global network-wide flight delay prediction capabilities, this undoubtedly leads to elevated deployment costs and time-consuming training, as well as an increased risk of data leakage and corruption. In recent years, federated learning (FL), as a novel distributed machine learning paradigm proposed by Google in 2016, has experienced notable advancements and is now beginning to take shape as a fresh strategy for tackling the above challenges [18]. The federated learning process involves conducting local learning model computations in the edge cloud and subsequently transmitting the local learning model parameters to the central cloud server for global model aggregation. This approach allows for the avoidance of direct interaction with the raw information, thereby enabling the attainment of a trade-off between model performance, communication overhead, and data security. In view of the advantages of federated learning, it is logical to fuse federated learning with deep learning to produce a powerful hybrid federated deep learning model, which offers many benefits and distinctive characteristics, including its capacity to facilitate a global server, address challenges associated with data collection, preserve privacy, enable high-performance computing, and allow for practical data customization. However, the hybrid federated deep learning model by far has not been applied to address the network-wide flight delay prediction problem. Thus, it is imperative to develop a customized hybrid federated deep learning model that is appropriate for network-wide flight delay prediction.

1.1. The main objective of study

To fill the gap, our research objective aims to train a robust central model based only on small-scale interactions with raw data (e.g., the interaction among airports in the same sub-region rather than global data interaction), which can predict flight delays across the entire network of airports. To this end, we propose a novel hybrid federated deep learning model (HFDL) for flight delay prediction at the network level that integrates federated learning and deep learning into a holistic predictive architecture. Specifically, the local spatial-temporal deep learning model is trained on each sub-region to capture spatial-temporal dependencies, while the federated learning fusion strategy is designed to update the global parameters aggregated from the local deep learning models. Finally, the trained global model is used to predict flight delays at the network level. In particular, the main contributions of this study are listed as follows:

- (1) We develop the HFDL model, which extracts the complex spatial dependencies of flight delays through DGCN and captures the temporal dependencies through RGRU. HFDL can be utilized to train a powerful central model with global flight delay prediction capabilities without the interaction of all sub-region raw data.
- (2) We design a local sub-region division strategy using the Louvain algorithm to partition the entire airport network into several local sub-regions so as to handle the locality of spatial dependency and improve the efficiency of the HFDL model.
- (3) To evaluate the performance of the proposed model, it is compared with the existing models by using a real-world dataset from the U.S. Bureau of Transportation Statistics (BTS) database, and the experimental results demonstrate that the proposed HFDL model performs better and has strong generalization and robustness.

The remainder of this paper is organized as follows. Section 2 provides an overview of the recent works pertaining to flight delay prediction. Section 3 introduces the problem specification and federated learning. Section 4 describes the proposed model in detail. Section 5 demonstrates the procedure of the experiments and analyzes the experimental results. Finally, Section 6 concludes the paper.

2. Literature review

Accurate flight delay prediction is the prerequisite for implementing efficient strategies for air traffic control. The findings of our survey indicate that a significant portion of early works use operational research, probabilistic models and statistical analysis in order to assess the likelihood of flight delays, or more precisely, the distribution of delays in flight schedules, where operational research uses optimization techniques, mathematical simulation, or queue theory to model air traffic operations [3,19–22]. These studies developed mathematical models based on historical flight data to predict flight delays. Notwithstanding, these models usually relied on some simplifying assumptions, ignored the complex correlations between flights, and had restricted capability to fit nonlinear relationships, leading to limitations in practical applications.

With the rapid development of artificial intelligence techniques, machine learning has gained significant focus as the increased availability of large-scale air traffic data [23–26]. For instance, Chen et al. employed a multi-label random forest classification approach in conjunction with air traffic operations to forecast the individual flight delays [27]. Guo et al. proposed a novel hybrid machine learning method integrating random forest regression with the Maximal Information Coefficient for predicting flight delay [28]. These models are able to better capture the complex relationships between data and improve prediction accuracy compared to mathematical statistical models. Yet, these models still have some limitations on effectively

utilizing the temporal correlation information of flight delays, which may lead to imprecise prediction results.

Recently, significant developments in deep learning techniques have accelerated the development of conventional transportation issues [29–32]. In contrast to conventional statistical models and machine learning techniques, deep learning exhibits superior capability in extracting intricate non-linear patterns. As flight delay data has strong nonlinearity and uncertainty, fortunately, neural networks have robust immunity to interference and powerful nonlinear mapping capabilities and can learn patterns of data changes under large-scale data inputs. For instance, models such as convolutional neural network (CNN) [33], recurrent neural network (RNN) [16] and long short-term memory network (LSTM) [34] are widely used in flight delay prediction. Nevertheless, the above efforts cannot address the complex spatial dependencies of flight delays caused by the interconnectedness of airports across various regions, which obligates researchers to propose more complicated or hybrid deep learning models to capture such spatial–temporal dependencies, namely network-wide flight delay prediction. In recent years, hybrid deep learning models based on graph convolutional (GCN) network have gained extensive development [17,31,35–38]. Some researchers have also begun to present hybrid models for the purpose of predicting network-wide flight delays. For example, [39] devised a hybrid deep learning model for spatial–temporal prediction of flight delay [39]. The results indicated that the utilization of a hybrid deep learning approach based on GCN can hold significant promise in addressing the issue of predicting flight delays. However, the studies mentioned above employ all available data for joint modeling without taking into account factors such as communication cost and data security. Lately, the utilization of federated learning (FL) has gained prominence as the means to evaluate data that is scattered across multiple sources, mostly due to its inherent capabilities in preserving privacy.

Federated learning is able to fuse heterogeneous information from multiple participants to collaboratively develop a shared predictive model by using the parameters of a locally learned model while remaining localized to the original training data [40]. The concept of federated learning was first introduced by Google in 2016, which is a novel distributed machine learning technique idea [18,41]. Due to the significant advancements in artificial intelligence and deep learning techniques in recent years, federated learning has emerged as a prominent area of research within the deep learning domain. This cutting-edge approach has garnered considerable attention and has yielded numerous noteworthy scientific research effects in the field of traffic prediction [42–44]. [45] applied FL into GRUs to model traffic speed and employed the cloud center server to fuse local models based upon data collected from the Caltrans Performance Measurement System database [45]. The result validated the efficacy of the proposed approach in generating reliable traffic predictions while upholding the integrity of data privacy. Moreover, federated learning also plays excellent capabilities in other fields as well, such as smart medical [46], natural language processing-related predictions [47], and recommendation systems [48].

To summarize, the findings in previous studies have yielded positive returns and provided meritorious insights into the spatial–temporal forecasting of flight delay. Yet, there are fewer studies that consider the use of federated learning for flight delay prediction at the network level, and the generalization performance of the existing model in the new environment remains unknowable. In this study, we predict flight delays at the network level by designing a novel hybrid deep learning model based on federated learning, namely HFDDL, which partitions the global airport network into several local sub-regions and further constructs the local spatial–temporal deep learning model. The proposed model enables global flight delay prediction without the interaction of raw data in sub-regions.

Table 1

Explanation of abbreviations.

Abbreviation	Explanation
DGCN	Diffusion Graph Convolutional Network
RGRU	Residual Gated Recurrent Unit
LSTM	Long Short-Term Memory
ACMS	Aircraft Condition Monitoring System
FL	Federated Learning
DCRNN	Diffusion Convolutional Recurrent Neural Network

3. Preliminaries

3.1. Problem specification

The network-wide delay forecasting problem studied in this paper is as follows. Consider that an airport network consists of N airports, and these airports are denoted as a weighted digraph $G = (V, E, A)$, where V is a set of nodes $|V| = N$, E is a set of weighted edges, and $A \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix. If airport i connects airport j through flights, then the matrix element A_{ij} of A is 1, and otherwise it is 0. At each time point t , the network G holds a feature matrix $\mathcal{X}^t \in \mathbb{R}^{N \times d}$. Given the feature matrix of the network G and T historical observations, the objective is to forecast the feature matrix (namely delays) of the network G in the future T' time steps, which can be formulated as follows:

$$[\mathcal{X}^{(t-T+1):t}, G] \xrightarrow{f} \mathcal{X}^{(t+1):(t+T')}, \quad (1)$$

where $\mathcal{X}^{(t-T+1):t} \in \mathbb{R}^{N \times d \times T}$ is the input of the model and f is the learned mapping function.

An explanation of some of the abbreviations that appear in the study is shown in the following Table 1.

3.2. Procedure of federated learning technique

FL is a burgeoning distributed machine learning approach. FL enables diverse participants to engage in the training of a cloud-based model without the need to transmit their individual information to the cloud. With the aid of FL, we can efficaciously mitigate the effects of network instability and circumvent the need for extensive data transmission when conducting model training. Consequently, the amount of time required for training is reduced. To be specific, the training procedures of FL at time $t, t \in \{1, 2, \dots, T\}$ are outlined as follows:

- **Initialization:** The participants need to enroll themselves with the cloud center server in order to partake in this process. The cloud center server employs a selection process to identify suitable participants who possess robust computing and communication capabilities to effectively execute the local model training tasks. Then the cloud center server sends an initialized global model θ^g to each participant.
- **Local training:** Each participant $k(k \in \{1, 2, \dots, K\})$ trains its local model with the local dataset and parameters $\theta_k(t)$. The objective of each participant is to tackle the following problem:

$$\arg \min_{\theta \in \mathbb{R}} L_k(\theta), \quad L_k(\theta) = \frac{1}{D_k} \sum_{i \in I_k} l_i(\theta), \quad (2)$$

where D_k represents the size of I_k , θ are the parameters of local model, and $l_i(\theta)$ is a local objective function.

- **Parameter aggregation:** The parameter aggregation of the FL is achieved by using FedAVG algorithm [41], i.e.,

$$\theta(t+1) = \frac{1}{K} \sum_{k=1}^K \theta_k(t+1). \quad (3)$$

Finally, the cloud center server distributes the updated global model $\theta(t+1)$ to all participants for the subsequent round of global model training.

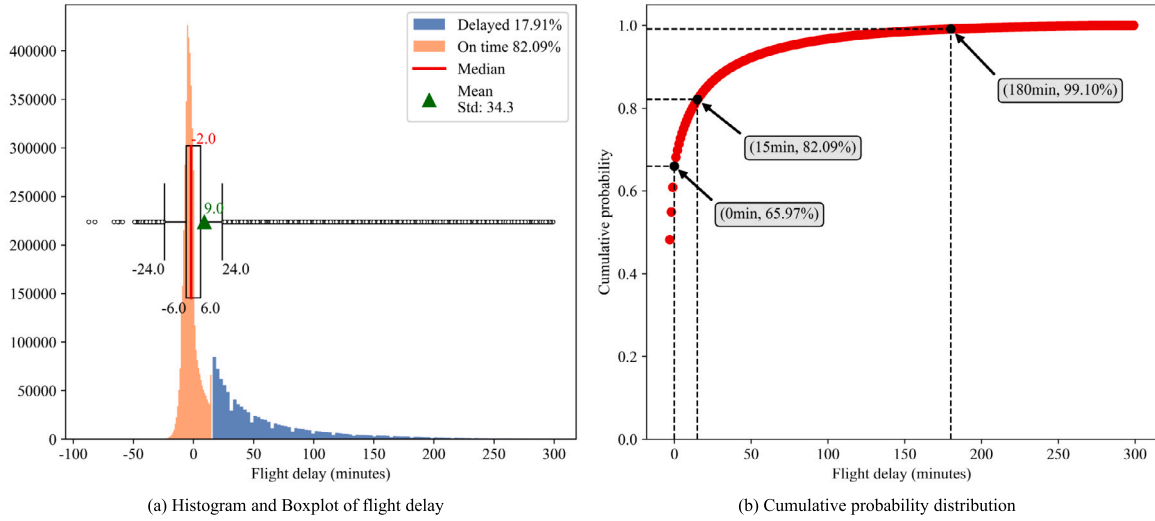


Fig. 1. Detailed statistics of historical flight delay.

4. Data and methodology

4.1. Dataset description

To conduct the network-wide flight delay prediction, we collect the experimental dataset from the U.S. BTS database.¹ The Bureau of Transportation Statistics is leading the way in the implementation of open data policies. As part of this effort, it has made available to the public extensive records of individual flights within the National Airspace System (NAS) spanning several years. This provision of data has been of immense benefit to researchers studying flight delays, affording them greater ease and opportunities in their investigations. In our study, we select flight records from May 1st, 2015 to December 31st, 2019 to form our case dataset. The original dataset comprises a total of 12,339,771 flights, which includes 373 airports and 6537 origin–destination (OD) pairs, most of which have less than one flight per day. Given the scope of the investigation, which centers on forecasting flight delays at the network level, we have opted to exclude OD pairs with less than 10 flights and airports with an average of less than 30 flights per day. Following the completion of data preprocessing, a comprehensive set of 52 airports and 392 OD pairs are identified and selected as the focal points of analysis for the network-wide flight delay prediction.

Fig. 1 exhibits the detailed statistics of the flight delay in our study dataset. Note that flight delays can take negative values, meaning that the aircraft left the gate early or arrived at the gate early. Using a very strict definition of on-time performance: delay ≤ 0 min, only 65.97% of flights are on time, while the rest are delayed by at least one minute. According to the universal flight delay rule: delay ≥ 15 min, the percentage of on-time flights reaches 82.09%, and 99.10% of flights are delayed within 180 min. The challenge in predicting flight delays lies not in the average delay magnitude but rather in the significant variability, as indicated by a standard deviation of 34.3 min.

In deep learning, we usually divide the dataset into a training set, a validation set, and a test set, where the training set is used to train the model, the validation set is used to adjust the parameters and structure of the model, and the test set is used to ultimately evaluate the performance of the model. Nonetheless, there may be some problems with this division method, such as it may lead to the model overfitting to the training set or not being able to fit the unknown data well. Therefore, we use cross-validation techniques to avoid this issue. As flight delay has temporal dependency and the conventional

Table 2

The statistical description of the divided dataset.

Property	1-fold	2-fold	3-fold	4-fold
Num.	2.352E6	3.528E6	4.704E6	5.88E6
Min	0.0000	0.0000	0.0000	0.0000
Max	73.2115	79.8462	79.8462	79.8462
Mean	14.7268	13.6195	12.4447	11.9962
St.deviation	13.0268	12.6359	11.9212	11.4070
Median	11.1250	9.9038	8.9615	8.8365
Skewness	1.3889	1.5795	1.7161	1.7608
Kurtosis	1.9308	2.7556	3.4801	3.8649

cross-validation method cannot maintain this property, we apply a 4-fold time-series split technique [49] to divide the dataset, which can maintain the temporal dependency. Time-series split method is a variant of K-Fold cross-validation [50], which first returns K folds into the training set and the (k+1)th fold as the validation set. Note that unlike standard cross-validation methods, consecutive training sets are supersets before them. In addition, it adds all remaining data to the first training partition, which is always used to train the model. The statistical description of the divided dataset is shown in Table 2. The model will undergo training on each of the divided dataset, and upon completion of training, the results will be averaged to yield the ultimate performance of the model. Moreover, considering the differences in data characteristics, data normalization is indispensable. We apply standard normalization to eradicate magnitude differences, which can be expressed as

$$\tilde{x} = \frac{x - \mu(x)}{\sigma(x)}, \quad (4)$$

where $\mu(x)$ and $\sigma(x)$ are the mean value and standard deviation of the samples, respectively.

4.2. Methodology: HFDL model

Deep learning models with graph-based structures can be computationally intensive and require significant time for model training. This is mainly due to the fact that these models need to deal with a large number of nodes and edges, resulting in a higher demand for computational resources. In addition, the training time of the models may also be longer, especially when training on large-scale datasets. To address these challenges, we adopt some strategies to reduce the complexity and training time of the models in our research. First, we employ the Louvain algorithm to divide the airport network into

¹ https://www.transtats.bts.gov/DL_SelectFields.asp?gnoyr_VQ=

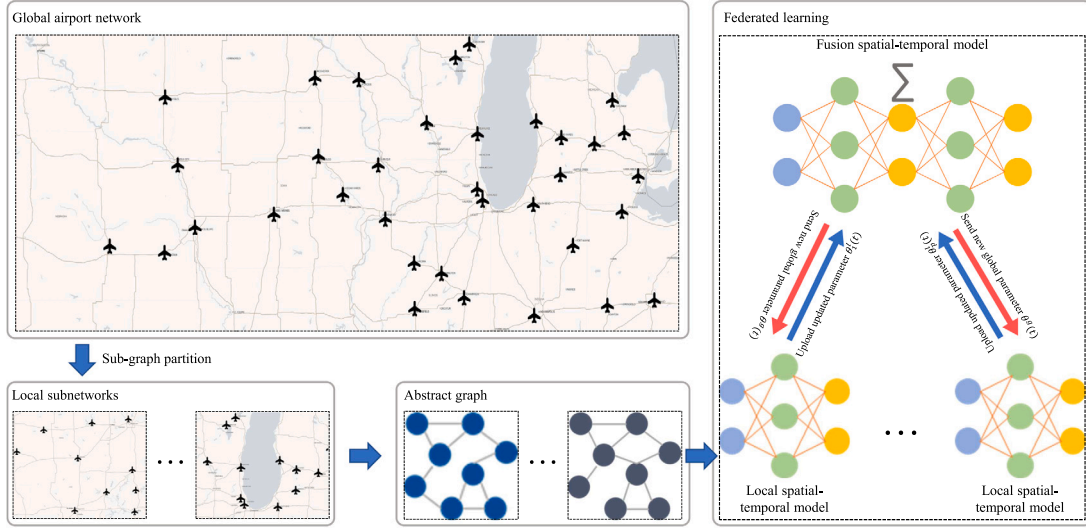


Fig. 2. The framework of the proposed HFDD model.

Table 3

Summary of notations.

Notations	Definitions
M	The network modularity
s	The sum of the weights of all the edges
σ	The activation function
p_i	The sum of the edge weights connected by node i
a_i	The sub-region in which node i is located
D_o	The out-degree diagonal matrices
D_i	The in-degree diagonal matrices
A'_w	The weighted adjacent matrix of the local sub-region
$\delta(\cdot, \cdot)$	The indicator function
W_z, W_r, W_h	The weight matrices of the update gate, the reset gate and the candidate hidden state, respectively
U_z, U_r, U_h	The weight matrices of the hidden state at the previous time in the update gate, the reset gate and the candidate hidden state, respectively.
e_z, e_r, e_h	The bias.

several sub-regions, which reduces the size of the graph and thus reduces the complexity of the model. Second, we adopt efficient graph neural network structures such as diffusion graph convolutional network (DGCN) and residual gated recurrent unit (RGRU) to improve the computational efficiency of the model. In addition, we utilize the distributed computing power of federated learning, and parallel techniques to accelerate the model training process. Our experimental results demonstrate that our proposed model can achieve excellent prediction performance while improving training efficiency.

Fig. 2 demonstrates a novel HFDD model architecture for network-wide flight delay prediction, which contains sub-region division, local spatial-temporal learning and federation fusion. The objective of sub-region division is to handle the locality of spatial dependency and improve the efficiency of the HFDD model. Then, a local spatial-temporal deep learning model is constructed for the divided sub-region. The federated learning model serves the purpose of aggregating the global parameters of each local model. Once the training of the global model is finished, it will be applied to the prediction tasks. A comprehensive elucidation of each constituent of the proposed HFDD model will be provided in the subsequent sections. The notations used in this study are summarized in Table 3.

4.3. Local sub-region division

The flight delay data exhibits a discernible sense of locality as evidenced by its spatial correlation. For instance, in a particular region, the airport network structure exerts a significant impact on the

interconnectivity of the traffic nodes, resulting in a robust level of interaction. The GCN-based method possesses the capacity to efficiently capture and depict the spatial correlation inherent in the network. Nevertheless, in cases where the network encompasses a vast expanse, GCN-based method necessitates a huge adjacency matrix to depict the network topology, which will result in the model consuming a greater amount of computational resources and increasing model training time. Furthermore, the act of consolidating the diverse node data also renders the model more susceptible to oversmoothing. Consequently, we usually employ the federated learning approach to train a forecasting model. The conventional approach to federated learning disregards the spatial correlation among nodes, as each node is treated as a client during the model training process [45]. In contrast to the presently employed centralized model, its predictive efficacy is suboptimal.

To address the aforementioned issues, we employ the Louvain algorithm [51] to partition the global spatial graph of the airport network into several sub-regions. The Louvain algorithm is a clustering technique that operates in a hierarchical manner and aims to optimize the modularity of a given network. The calculation of modularity is performed in the following manner:

$$\begin{cases} M = \frac{1}{2s} \sum_{i,j} \left[A_{ij} - \frac{p_i p_j}{2s} \right] \delta(a_i, a_j) \\ s = \frac{1}{2} \sum_{i,j} A_{ij} \end{cases} \quad (5)$$

where M denotes the network modularity, p_i represents the sum of the edge weights connected by node i , and a_i is the sub-region in which node i is located. $\delta(a_i, a_j)$ signifies a indicator function, i.e., if both nodes are part of the identical sub-region, then $\delta(a_i, a_j)$ equals 1; otherwise, $\delta(a_i, a_j)$ equals 0.

4.4. Personalized spatial-temporal learning mechanism

To model the evident spatial-temporal characteristics of network-wide flight delays, this section introduces a novel personalized spatial-temporal (ST) learning mechanism within each local sub-region by stacking multiple ST cells and a fully connected layer to constitute a deep structure, as shown in Fig. 3. Each ST cell contains a diffusion GCN unit and a residual GRU unit, where the diffusion GCN unit is employed to capture topological structure of airport network to extract the spatial dependency information and the obtained time series with spatial dependency are further inputted into the residual GRU unit to capture the temporal dependency of flight delays. Finally, we fuse the obtained spatial-temporal characteristics to predict the state of flight delays using the fully connected layer. Subsequently, we will elaborate on spatial dependency modeling and temporal dependency modeling.

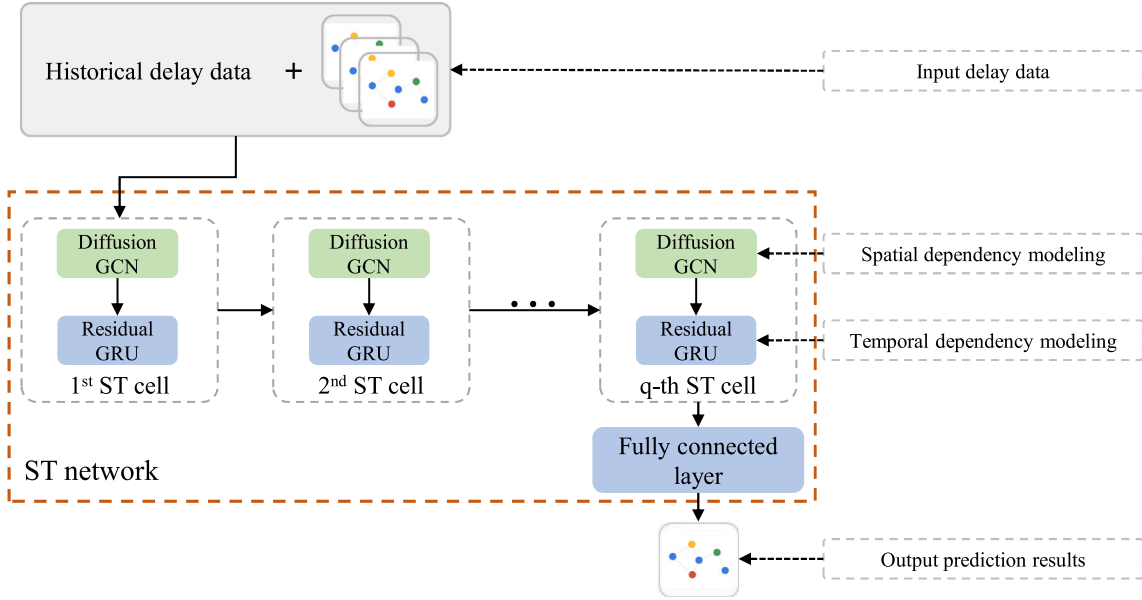


Fig. 3. The structure of the local spatial-temporal deep learning model.

4.4.1. Diffusion GCN for spatial dependency modeling

Flight delays exhibit a complex spatial dependency due to the interconnectedness of airports across various regions. The conventional convolutional neural network (CNN) is capable of extracting localized spatial characteristics, albeit limited to Euclidean space, such as images or regular grids [52]. The airport network exhibits a graph-like structure rather than a two-dimensional grid, rendering the CNN model inadequate in capturing the intricate topological characteristics of the airport network. Consequently, the CNN model fails to accurately grab spatial dependencies. Recently, there has been a significant focus on extending CNN to GCN. The GCN has the ability to effectively process data with arbitrary graph structures, which has garnered considerable interest from researchers and practitioners alike. The GCN model utilizes a Fourier domain filter to operate on the nodes of a graph and their immediate neighboring nodes, where the filter is designed to capture spatial characteristics between the nodes. By stacking multiple convolutional layers, the GCN model can be constructed. Nevertheless, the occurrence of flight delays at airports can have a substantial influence on upstream and downstream airports, thereby highlighting a limitation on efficaciously addressing this particular phenomenon using GCN. For this purpose, we introduce the diffusion GCN to model the bidirectional diffusion process, which allows the model to accurately capture the impact of upstream and downstream delays. Specifically, the diffusion GCN can be expressed as

$$X_{t \times Gf_{\theta_d^l}} = \sum_{b=0}^{B-1} \left(\theta_{b,1}^l (\mathbf{D}_o^{-1} A_w^l)^b + \theta_{b,2}^l (\mathbf{D}_I^{-1} (A_w^l)^T)^b \right) x_t, \quad t = 1, \dots, T_a \quad (6)$$

$$\mathbf{DGC} = \sigma \left(\sum_{t=1}^{T_a} X_{t \times Gf_{\theta_d^l}} \right), \quad t' = 1, \dots, T'_a \quad (7)$$

where $\mathbf{D}_o$ and $\mathbf{D}_I$ represent the out-degree and in-degree diagonal matrices respectively, and $\theta_{b,1}^l \in \mathbb{R}^B$ and $\theta_{b,2}^l \in \mathbb{R}^B$ denote the corresponding weight matrices. A_w^l is the weighted adjacent matrix of the local sub-region. $\theta_d^l \in \mathbb{R}^{B \times 2}$ are the parameters for the filter while $(\mathbf{D}_o^{-1} A_w^l)$ and $(\mathbf{D}_I^{-1} (A_w^l)^T)$ denote the transition matrices of the diffusion process and the reverse one, respectively. The results of Eqs. (6) and (7) represent spatial characteristics extracted from b -hop neighborhoods and the output of diffusion GCN, respectively. σ signifies the activation function.

4.4.2. Residual GRU for temporal dependency modeling

The issue of temporal dependency poses a significant challenge in the field of flight delay forecasting. Currently, the RNN stands as the predominant neural network model employed for the purpose of processing sequence data and has demonstrated efficacy in grabbing the temporal dependencies among variables when modeling flight delay data [34]. Nevertheless, the conventional RNN is constrained in its predictive capabilities due to inherent defects such as gradient disappearance and explosion [53]. The GRU model [54] as the variant of RNN has been proven to tackle the above issues. GRU contains the update gate z_t , the reset gate r_t , and the hidden state h_t . The settings of gate mechanisms can effectively help to address the defects of RNN. To be specific, the formula of GRU procedure is expressed as follows:

$$z_t = \sigma(W_z \times x_t + U_z \times h_{t-1} + \epsilon_z) \quad (8)$$

$$r_t = \sigma(W_r \times x_t + U_r \times h_{t-1} + \epsilon_r) \quad (9)$$

$$\tilde{h}_t = \tanh(W_h \times x_t + U_h \times (r_t * h_{t-1}) + \epsilon_h) \quad (10)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (11)$$

where W_z , W_r , and W_h denote the weight matrices of the update gate, the reset gate and the candidate hidden state respectively while ϵ_z , ϵ_r , and ϵ_h represent the corresponding bias. Additionally, U_z , U_r , and U_h stand for the weight matrices of the hidden state at the previous time h_{t-1} in the update gate, the reset gate and the candidate hidden state, respectively.

In order to fully extract the temporal features of sequence data, we propose a residual GRU, motivated by the residual network [55], which modifies the original GRU by incorporating an identity mapping through a shortcut path (see red arrow in Fig. 4). A residual GRU unit has the same first three formulas of z_t , r_t , and $\tilde{h}_t$, but h_t is modified as

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t + W_{trans} x_t \quad (12)$$

where the additional term $W_{trans} x_t$ functions as an identity mapping, whereby it directly transfers the linear projection of the input x_t to the output through a shortcut.

Due to the ST network stacks multiple ST cells for spatial-temporal modeling and each ST cell contains a diffusion GCN unit and a residual GRU unit, thereby we formulize the ST network in the following. Suppose that the input of the q th layer is a sequence data denoted as

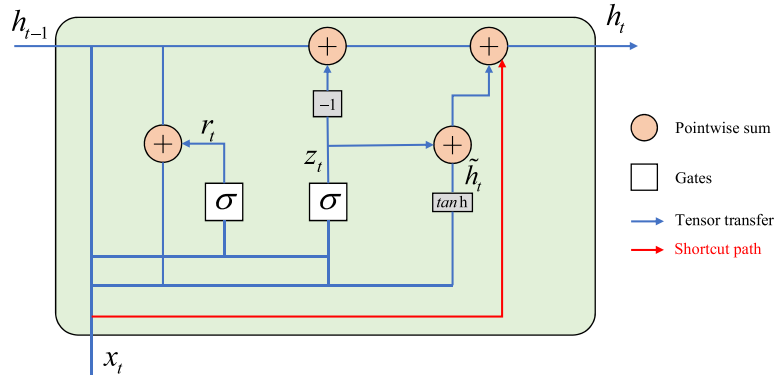


Fig. 4. The inner structure of a residual GRU unit, which has one more shortcut path (red arrow) than an original GRU unit.

$X^q = (X_0^q, X_1^q, \dots, X_{T_s-1}^q)$ with $X_t^q \in \mathbb{R}^N$, where T_s denote the number of time steps and N is the number of sub-region nodes. The formula of a ST network unit at the time step t in the q th layer is given as follows:

$$z_t^q = \sigma(W_z^q * x_t^q + U_z^q * h_{t-1}^q + \epsilon_z^q) \quad (13)$$

$$r_t^q = \sigma(W_r^q * x_t^q + U_r^q * h_{t-1}^q + \epsilon_r^q) \quad (14)$$

$$\tilde{h}_t^q = \tanh(W_h^q * x_t^q + U_h^q * h_{t-1}^q \times (r_t^q * h_{t-1}^q) + \epsilon_h^q) \quad (15)$$

$$h_t^q = (1 - z_t^q) * h_{t-1}^q + z_t^q * \tilde{h}_t^q + W_{trans}^q * x_t^q \quad (16)$$

where $*_G$ denotes the graph convolution operator.

4.4.3. Attention mechanism

While residual GRU can learn the long-term dependencies of the inputs, the outputs of the hidden states are of different importance to the final output. Recently, [56] introduced the concept of attention mechanism in deep learning, where different attention is given to each part of the input sequence [56]. Introducing the attention mechanism can enhance the residual GRU model's ability to model the importance of different time steps in the input sequence, further improving the model's expressive and generalization abilities. By adaptively adjusting the weights of each time step, the attention mechanism can help the model better capture crucial information. Specifically, the attention mechanism assigns the weight w_i for hidden state h_i as follows:

$$w_i = \frac{\exp(\tanh(w_A h_i))}{\sum_{i=1}^{L_x} \exp(\tanh(w_A h_i))} \quad (17)$$

where w_A indicates a trainable weight matrix, $\tanh(\cdot)$ indicates the attention function and L_x indicates the length of input sequence.

Subsequently, the attention vector H is produced as the result of the attention mechanism, whereby it amalgamates the output of each hidden layer with its respective weight.

$$H = \sum_{i=1}^{L_x} w_i h_i. \quad (18)$$

4.5. Federated learning fusion strategy

The centralized deep learning model necessitates the transmission of flight delay data from airports dispersed across the airport network to the cloud center server, resulting in a substantial communication burden. Additionally, the time needed for transmitting an enormous amount of flight delay data will significantly impact the training period for the model. The conventional federated learning algorithms are unable to exploit the spatial patterns, leading to inferior prediction accuracy when compared to the current deep learning model. For this purpose, we present a novel federated learning fusion strategy, which includes the following procedures.

STEP 1. The process of partitioning the airport network into multiple sub-regions can be achieved by applying the local sub-region division discussed in the preceding subsection. It can be expressed as $G = \{G_1, G_2, \dots, G_p\}$, where p denotes p sub-regions. In each sub-region, we select a representative node v_i and then a local spatial-temporal model is generated on this representative node. The local spatial-temporal model has two parameters: local parameters θ^l and global parameters θ^g .

STEP 2. The cloud center server selectively designates some nodes from the representative node-set $\mathcal{V}$ where such nodes are called volunteer nodes, to participate in current round of training and subsequently sends the global parameters θ^g to the volunteer nodes. It should be noted that the local parameters θ^l are exclusively situated within the local model. At the onset of the training process, the cloud center server sends an initialized global model θ^g to each representative node, i.e., $\theta^l \leftarrow \theta^g$. The representative node further utilizes the previously acquired parameters for subsequent training processes.

STEP 3. The ADAM method [57] is employed to train the local spatial-temporal model and the model parameters θ^l and θ^g are updated. The training process exclusively utilizes the locally available data within the sub-region. Specifically, the training process can be represented as follows

$$\begin{cases} \theta_t^l \leftarrow \theta_t^g \\ \theta_{t+1}^l \leftarrow \text{LocalUpdate}(\tau, v_i, G_i, \theta_t^l) \end{cases} \quad (19)$$

where τ represents the coefficient of the soft update. The updated process can be implemented in parallel on each volunteer node.

STEP 4. The cloud center server collects and aggregates the local parameters from all volunteer nodes, subsequently updating them to the global parameters of the cloud model. The aggregation process is achieved by using Eq. (20).

$$\theta_{t+1}^g = \frac{1}{p} \sum_{i=1}^p (\theta_{i,t+1}^l + \epsilon_i) \quad (20)$$

where ϵ_i indicates the gaussian noise added to i th model parameters.

The aforementioned learning process is iterative and will continue to repeat STEPS 2 – 4 until the global model converged. From STEP 1 to STEP 4, they are the core mechanism of HFDL. The utilization of HFDL demonstrates its effectiveness in decreasing the computational time of the model and mitigating the burden of communication, while maintaining a commendable level of accuracy in the prediction task.

4.6. Model performance evaluation indices

The qualification and assessment of the proposed networks represent the primary stages involved in the procedure for generating the model. Recognizing the validity and sufficiency of the simulations' findings for the selected objectives is crucial. The utility of prediction is assessed by evaluating the dependability of the training, testing, and all

data stages of the previous methodologies. To this end, we select several widely-used modeling utility statistical criteria to evaluate the performance and precision of the models, namely coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), the variance accounted factor (VAF), performance index (PI), scatter index (SI), where the smaller values of MAE, PI, RMSE and SI are proper while the larger values of VAF and R^2 are more reliable [58–60]. The specific metrics are provided as follows:

$$R^2 = \left(\frac{\sum_{q=1}^Q (x_q - \bar{x})(z_q - \bar{z})}{\sqrt{\left[\sum_{q=1}^Q (x_q - \bar{x})^2\right] \left[\sum_{q=1}^Q (z_q - \bar{z})^2\right]}} \right)^2, \quad (21)$$

$$RMSE = \sqrt{\frac{1}{Q} \sum_{q=1}^Q (z_q - x_q)^2}, \quad (22)$$

$$MAE = \frac{1}{Q} \sum_{q=1}^Q |z_q - x_q|, \quad (23)$$

$$VAF = \left(1 - \frac{\text{var}(x_q - z_q)}{\text{var}(x_q)} \right) * 100, \quad (24)$$

$$PI = \frac{1}{|\bar{x}|} \frac{RMSE}{\sqrt{R^2 + 1}}, \quad (25)$$

$$SI = \frac{RMSE}{\bar{x}}, \quad (26)$$

where $z_q, x_q, \bar{x}, \bar{z}$, denote the prediction value, observed value, average of observed value, average of prediction value, respectively. Q indicates the number of samples. Regarding SI, $SI < 0.05$ indicates the “excellent precision”, $0.05 < SI < 0.1$ indicates the “good precision”, $0.1 < SI < 0.15$ indicates the “fair precision”, and $SI > 0.15$ indicates the “poor precision”.

Furthermore, the $t_{statistic}(T_{stat})$, which measures the significance between the output and estimated values, is also used to acquire further information on suggested models’ performance utilizing uncertainty with a 95% confidence level (U_{95}) [61]:

$$T_{stat} = \frac{1}{\bar{x}} \sqrt{\frac{(Q-1)MBE^2}{RMSE^2 - MBE^2}}, \quad (27)$$

$$U_{95} = 1.96 \sqrt{(SD^2 + RMSE^2)}, \quad (28)$$

where $MBE = \frac{1}{Q} \sum_{q=1}^Q (z_q - x_q)$ and the value 1.96 represents the coverage factor for a sizeable amount of data points at a 95% confidence level. SD refers to the standard deviation of the difference between the estimated and observed data. The closer the values of these two metrics are to 0, the better the model performance.

5. Results and discussion

5.1. Hyperparameter setting

In order to verify the superior performance of the proposed model, various models are compared for network-wide flight delay prediction. The comparative models are presented below: LSTM [34], T-GCN [62], DCRNN [63], FedGRU [45]. As the models contain multiple hyperparameters, we list these parameters in Table 4. These best hyperparameters are selected using the Tree-structured Parzen Estimator (TPE) algorithm [64]. All experimental comparisons are conducted on a desktop computer with 3.8 GHz Intel Core i7 10700K CPU, NVIDIA GeForce RTX 3050Ti GPU, and 64 GB of RAM, and all the training processes are implemented with Python 3.6 and Pytorch version 1.7. Figures are produced in Python and Origin2022 student version.

5.2. Analysis of precision and generalization

We train the LSTM [34], T-GCN [62], DCRNN [63], FedGRU [45] and HFDDL in the delay dataset. The convergence curves of RMSE and R^2 in the training process are shown in Fig. 5. As can be seen, after 100 epochs of training, the above models have achieved convergence. The curves obviously depict that the HFDDL model has better convergence rate than the other competitive models, having attained the smallest RMSE of 0.8416 at the minimum iteration of around 64 and the largest R^2 of 0.9989 at the iteration of 95. The convergence curves validate the effectiveness of the HFDDL model, which can be used to predict the network-wide flight delays. We then plot predicted results and real results of the HFDDL model in Fig. 6. Particularly, we randomly choose 4 sub-regions (#1, 2, 4, 5) and arbitrary 168 continuous timesteps (1 week as time step is 1 h) to compare the predicted value with the real value. In the airport network, the delays exhibited by local sub-regions demonstrate distinct patterns of variation. The figure illustrates that HFDDL exhibits the ability to adapt to the distinct flight delay data patterns observed in various sub-regions, while consistently achieving a high level of prediction accuracy.

Fig. 7 shows one-step ahead (in the one-hour interval) forecasting performance of the different models in the training and test phases under selected evaluation metrics. The specific values of R^2 , RMSE, MAE, VAF, PI and SI are given in Table 5, where the bold part implies that the HFDDL model has the best performance. From the results the prediction accuracy of HFDDL outperforms the other compared models under all evaluation metrics, with $R_{train}^2 = 0.8847$, $R_{test}^2 = 0.8532$, $RMSE_{train} = 6.1849$, $RMSE_{test} = 6.9755$, etc. Among the models in Table 5, LSTM [34], T-GCN [62], and DCRNN [63] are all centralized deep learning models, while FedGRU [45] and HFDDL are the federated learning models. In addition, from the test results in Table 5, the larger the data amount the more powerful the predictive performance of the trained model.

The SI values for the training and test phases are shown in Fig. 7(c) and (f). Regarding SI, a model includes an excellent precision (EP) with $SI < 0.05$, a good precision (GP) with $0.05 < SI < 0.1$, a fair precision (FP) with $0.1 < SI < 0.15$, and a poor precision (PP) with $SI > 0.15$. As can be observed in the training and test phase, the SI values in the LSTM [34], T-GCN [62], DCRNN [63] and FedGRU [45] models are more than 0.05 but below 0.1, classified in GP, with the exception of the proposed HFDDL model, which is classified in EP. In addition, the SI values of the models in the test phase are generally higher than those in the training phase, while the SI values of the HFDDL model proposed in this paper remain below 0.05 in the test phase (0.0496). Therefore, it is clear by carefully studying the SI values for different models in the training and test phases that the HFDDL model may be considered as the superior technique.

Furthermore, it can be noticed that HFDDL demonstrates significant advantages in terms of training time. In real-time flight delay forecasting tasks, the time expenditure is a critical measure. In contrast to HFDDL, the training time of alternative baseline model showcases a comparatively higher magnitude. The iterative calculation approach employed by DCRNN [63] necessitates a significant amount of time, leading to an extended computational period for the model. Despite being deployed in a similar manner to HFDDL and practicing a comparatively low time cost, FedGRU [45] remains inadequate for real-time prediction tasks as FedGRU does not model the spatial dependencies of flight delays.

HFDDL employs the local sub-region division strategy to partition the airport network into several sub-regions. Subsequently, the federated learning fusion strategy is applied to train the models. The model incorporates both the local spatial dependency and the global temporal dependency. Partitioning the global airport network into sub-regions serves to decrease the number of parameters associated with each submodel, while simultaneously mitigating the time costs associated

Table 4
The hyperparameter settings for each method.

Method	Parameters	Value	Method	Parameters	Value
LSTM	Number of hidden units	50	T-GCN	Number of hidden units	100
	Number of hidden layers	2		Number of hidden layers	3
	Batch size	64		Batch size	64
DCRNN	Number of hidden layers	2	HFDL	The historical time step	8
	Number of hidden units	64		The future time step	1
	Maximum steps of random walks	3		Number of hidden units	64
FedGRU	Batch size	128		Number of hidden layers	3
	Number of organizations	20		Batch size	32
	Number of hidden layers	3		Learning rate	0.01
	Number of hidden units	100		The communication rate	0.5

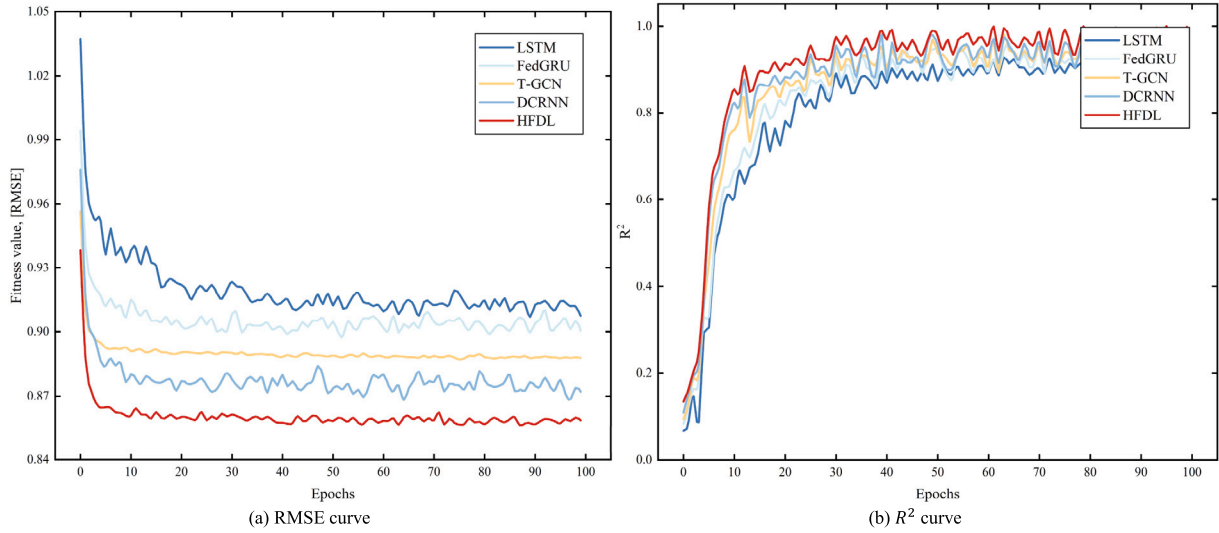


Fig. 5. The convergence curve of RMSE and R^2 .

Table 5
Average performance comparison of different models in the training and test phases.

	Model	R^2	RMSE	MAE	VAF	PI	SI	T_{stat}	U_{95}	Training time (s)
Training phase	LSTM	0.8216	6.4197	4.8616	82.1616	0.0563	0.0563	0.2073	17.7242	494.5
	T-GCN	0.8627	6.3291	4.8489	86.2743	0.0545	0.0544	0.7065	17.3425	256.2
	DCRNN	0.8693	6.3261	4.8015	86.9316	0.0533	0.0528	0.7730	17.2186	2957.9
	FedGRU	0.8573	6.3337	4.8531	85.7322	0.0552	0.0552	0.1535	17.4549	180.3
	HFDL	0.8847	6.1849	4.7340	88.4739	0.0526	0.0472	0.0599	17.0432	140.3
Test phase	LSTM	0.7962	7.6902	5.7192	79.6207	0.0578	0.0643	3.2676	21.3145	–
	T-GCN	0.8126	7.1928	5.4485	81.2637	0.0553	0.0624	1.2350	19.9308	–
	DCRNN	0.8253	7.1856	5.4321	82.5342	0.0542	0.0581	0.9100	19.9079	–
	FedGRU	0.8025	7.6069	5.6800	80.2513	0.0569	0.0636	3.2191	21.0847	–
	HFDL	0.8532	6.9755	5.3338	85.3216	0.0531	0.0496	0.1499	19.3350	–
	HFDL(1-fold)	0.8318	7.4083	5.3749	83.1772	0.0603	0.0507	0.1654	20.6771	–
	HFDL(2-fold)	0.8406	6.8937	5.3625	84.0619	0.0521	0.0501	0.1511	19.9089	–
	HFDL(3-fold)	0.8639	6.8332	5.3017	86.3961	0.0508	0.0491	0.1464	18.4913	–
	HFDL(4-fold)	0.8765	6.7668	5.2961	87.6512	0.0492	0.0485	0.1367	18.2627	–

The bold fonts signify the optimal results. HFDL(1-fold) indicates that the model is trained on the first cross-validate dataset (Table 2).

with transmitting a large volume of data. Consequently, the HFDL model exhibits comparatively reduced training time.

The values of U_{95} and T_{stat} as workability assessors in the training and test phases are depicted in Fig. 8. The smaller the uncertainty value, the stronger the generalization capacity, which implies that the model can be well applied to new regions. With $U_{95-train} = 17.0432$, and $U_{95-test} = 19.3350$, the HFDL scenario explicitly has the lowest amounts of uncertainty when compared to other models, proving its superior generalization ability. Moreover, the values of T_{stat} are somewhat identical to the trend of U_{95} . Same as U_{95} , the values of T_{stat} for HFDL model are lower than others in the training phase at 0.0599 and test phase at 0.1499, followed by DCRNN ($T_{stat-train} = 0.773$ and $T_{stat-test} = 0.91$), T-GCN ($T_{stat-train} = 0.7065$ and $T_{stat-test} = 1.235$), FedGRU ($T_{stat-train} =$

0.1535 and $T_{stat-test} = 3.2191$). Overall, the descriptions together suggest that HFDL could perform better than other competitors.

5.3. Sensitive analysis

In this section, we assess the sensitivity of the relevant parameters to the model. We use the quantities of three statistical performance criteria: $RMSE$, T_{stat} , U_{95} . The impact of soft update coefficient τ is displayed in Table 6. As can be seen that, when $\tau = 0.1$, the HFDL model obtains the highest performance. The smaller the value of τ , the better the performance. The RMSE at $\tau = 1.0$ is 11.1% higher than that at $\tau = 0.1$. The decrease in the value of τ results in a reduced retention of local information by the local model. Consequently, more emphasis

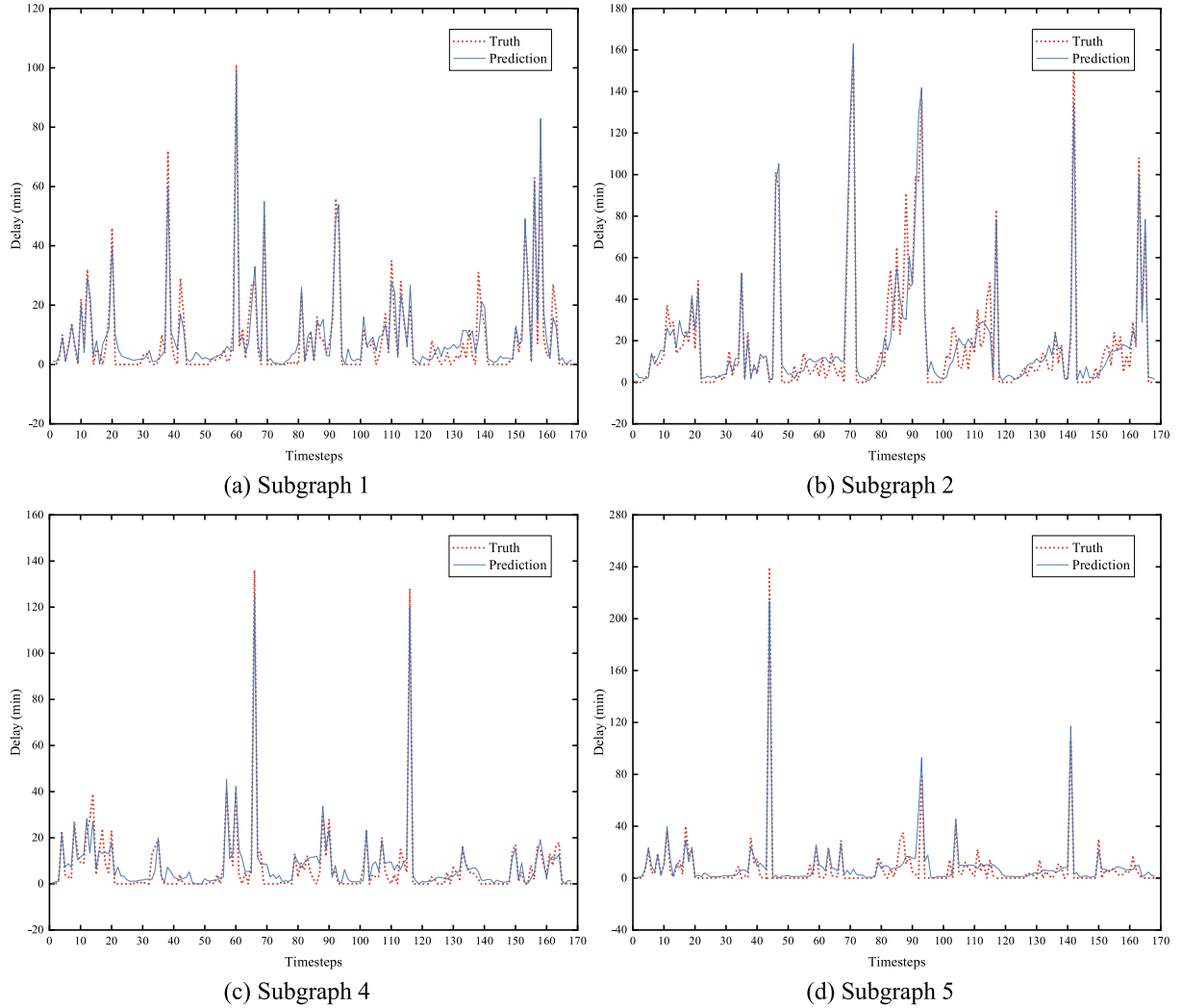


Fig. 6. Comparison of the predicted values and real values of 4 randomly selected sub-regions in 168 continuous timesteps.

Table 6

The impact of soft update coefficient τ to HFDDL model.

Model	Training phase			Test phase		
	RMSE	T_{stat}	U_{95}	RMSE	T_{stat}	U_{95}
0.1	6.3291	0.0599	17.5432	6.9755	0.1499	19.3350
0.2	7.1106	0.1202	19.7030	6.9790	1.3969	19.3445
0.3	7.1117	0.1685	19.7052	6.9794	1.4896	19.3455
0.4	7.1127	0.2103	19.7072	6.9798	1.5699	19.3464
0.5	7.1138	0.2553	19.7093	6.9803	1.6563	19.3475
0.6	7.1149	0.2944	19.7113	6.9807	1.7312	19.3484
0.7	7.1161	0.3400	19.7136	6.9812	1.8188	19.3495
0.8	7.1173	0.3854	19.7159	6.9818	1.9059	19.3508
0.9	7.1186	0.4320	19.7184	6.9825	1.9954	19.3521
1.0	7.1200	0.4786	19.7210	6.9832	2.0847	19.3534

Table 7

The sensitive analysis of communication rates γ .

Model	Training phase			Test phase		
	RMSE	T_{stat}	U_{95}	RMSE	T_{stat}	U_{95}
Centralized	6.6728	0.0665	18.0721	7.5682	0.1579	20.6317
HFDDL ($\gamma = 0.4$)	6.4797	0.0646	17.5492	7.0914	0.1707	19.5320
HFDDL ($\gamma = 0.5$)	6.3892	0.0637	17.5461	7.6671	0.1623	19.8977
HFDDL ($\gamma = 0.6$)	6.5944	0.0677	17.8597	7.6658	0.1658	20.9012
HFDDL ($\gamma = 0.7$)	6.6426	0.0662	17.9904	7.5342	0.1691	20.5390
HFDDL ($\gamma = 0.8$)	6.5159	0.0649	17.6473	7.5967	0.1703	20.7093
HFDDL ($\gamma = 0.9$)	6.4858	0.0646	17.5656	7.1617	0.1688	20.6704
HFDDL ($\gamma = 1.0$)	6.6969	0.0667	18.1375	7.4635	0.1736	20.8062

is placed on training the global model, leading to an enhancement in the performance of HFDDL.

Furthermore, the HFDDL model is utilized to generate multigroup predictions at varying communication rates γ . The evaluation results of the HFDDL model under each communication rate are compared with those of the centralized case. The comparison results are depicted in Table 7. The evaluation results obtained using various values of γ exhibit a significant level of consistency with the evaluation results observed in the centralized scenario. The performance of the HFDDL is comparable

to that of the centralized case, indicating that the federated model does not exhibit inferior performance compared to the centralized case, as suggested in [65].

Federated learning indeed brings about certain challenges, including communication overhead and synchronization issues. In our research, we have addressed these technical limitations and potential bottlenecks in the federated learning process, particularly when aggregating local models. To mitigate communication overhead, we have designed a hybrid federated deep learning model (HFDDL) that emphasizes updating model parameters only on the central server without accessing or sharing any private data. This approach significantly reduces the

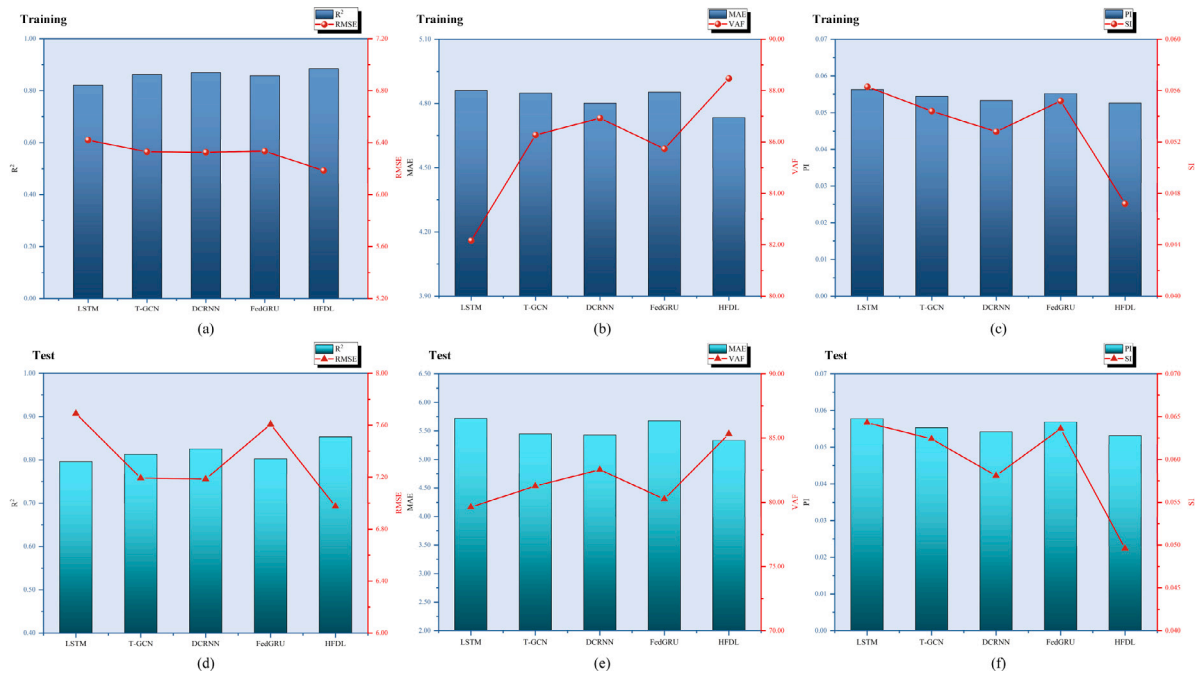


Fig. 7. The R^2 , RMSE, MAE, VAF, PI and SI values of models in the training and test phase.

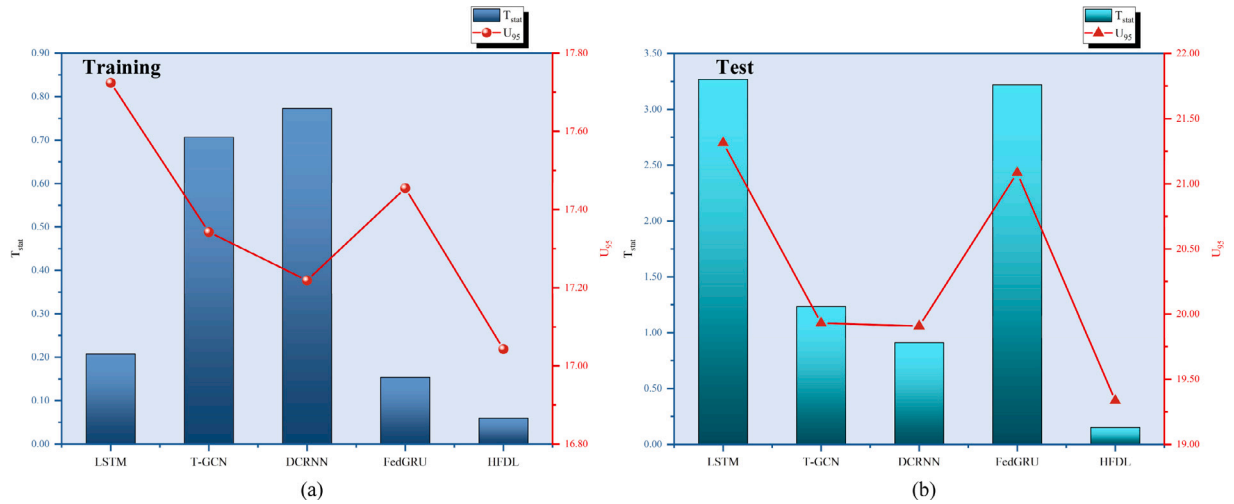


Fig. 8. The U_{95} and T_{stat} values of models in the training and test phase under p -value < 0.01.

need for extensive data communication between the central server and local devices, thereby alleviating the communication burden. Regarding synchronization issues, we have employed the Federated Averaging (FedAVG) method to aggregate local models. FedAVG ensures that the central model is updated based on the weighted average of the local models' parameters, allowing for synchronization while preserving the specific information captured by each local model.

5.4. Explainability analysis

After developing a model, it is imperative to determine the underlying mechanism by which the model generates predictions. This understanding is crucial for assessing the model's performance and identifying situations where human intervention is necessary. Outside of the interactive computing platform, such as Jupyter notebook, the importance of model explainability/interpretability becomes even more critical for decision-makers.

Attention mechanism plays an important role in deep learning models and has a positive impact on the explainability of the model [66]. The attention mechanism enables the model to pay more attention to specific parts of the input data, thus allowing the model to focus more on the information that has an impact on the prediction results. By visualizing the attention weights, it helps to understand how much attention the model pays to different input features and how important these features are to the final prediction results.

Fig. 9 depicts a visualization of the average temporal attention weights. The average attention weights of each looked-back step in the prediction process are computed across the entire test dataset. Based on the heat map, it is evident that the most recent periods hold the most importance for future prediction. The significance of each looked-back step can be quantified by the mean level of attention received, as shown in Fig. 9(b). The newest steps hold the most significance, but the relevance of subsequent periods tends to diminish as they move further away from the current step.

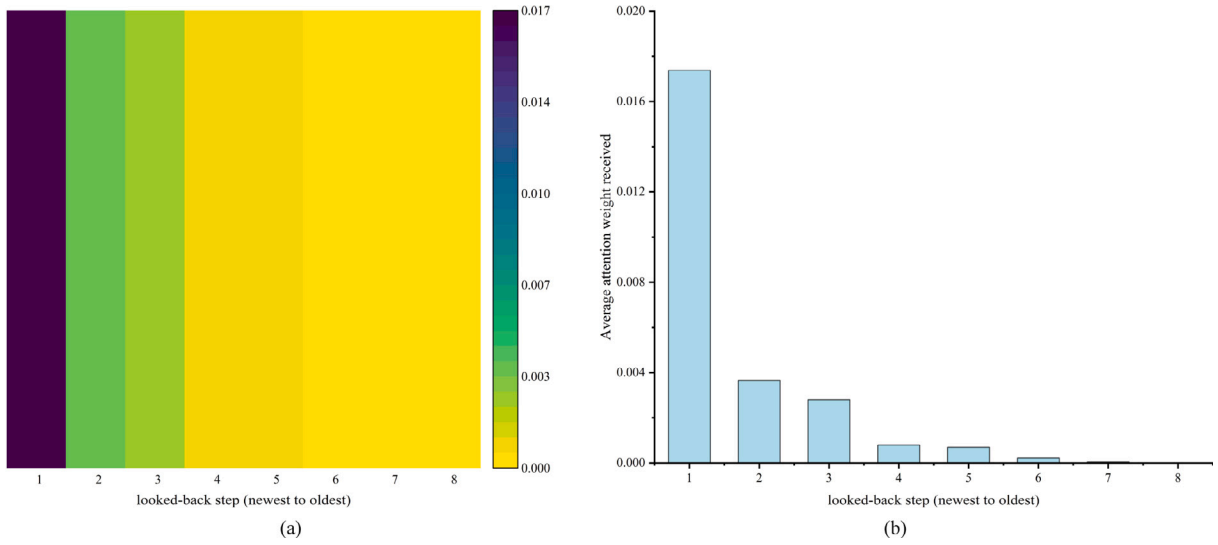


Fig. 9. Average temporal attention received by different looked-back steps.

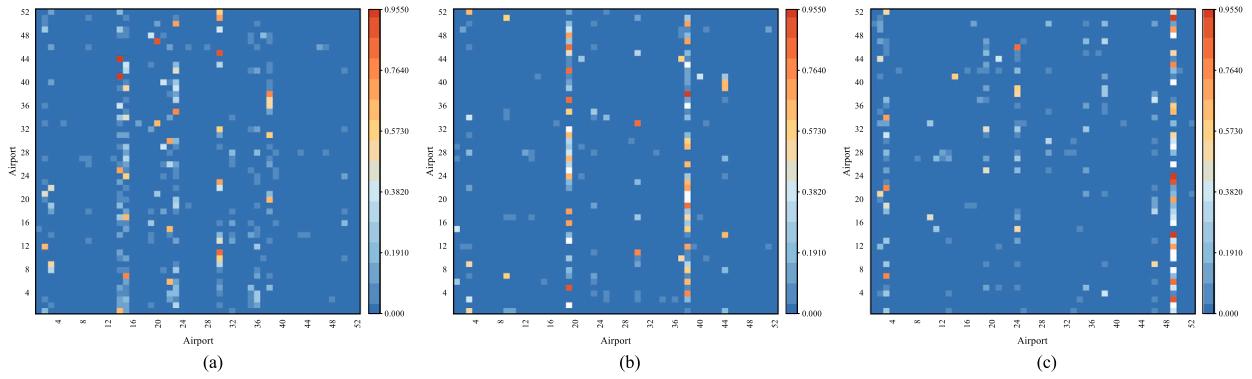


Fig. 10. Visualization of spatial-temporal dependencies from the perspective of global network.

The spatial dependencies are also visualized in this part. In order to intuitively feel the locality of spatial dependency in the global network, Fig. 10 randomly visualizes three different spatial dependencies from the perspective of global network in the form of heatmap, where the global network includes 52 airports. In the figure, the darker red indicates a stronger spatial dependencies. As can be seen, the spatial dependencies present significant locality, which is exactly the purpose of our local sub-region division strategy, so as to speed up the training of the model and reduce the consumption of computational resources. The visualization results demonstrate that the proposed model can effectively explore the spatial-temporal dependencies of flight delays, providing insightful suggestions for aviation authorities and airport regulators to formulate proactive air traffic control strategies for mitigating flight delays.

6. Conclusion and future work

This study first proposes a novel HFDL model to address the problem of flight delay prediction at the network level. We first divide the global graph of the airport network into multiple sub-regions, and then develop personalized local spatial-temporal deep learning models for each sub-region to extract the spatial-temporal dependencies of the sub-regions. The local models are aggregated through a federated learning fusion strategy to form a robust central model that removes the limitations of global data sharing and privacy guarantees. To evaluate the performance of the proposed model, it is compared with the existing

models by using a real-world dataset from the U.S. Bureau of Transportation Statistics (BTS) database, where the dataset are separated by the time series cross-validation technique. It is worth mentioning that the proposed model is created using some specific data, so the results are specialized and cannot be simply or arbitrarily generalized to all field scenarios. The main conclusions are given as follows:

- Findings present that all the compared models include acceptable performance in flight delay predictions among them the advantage of the proposed model HFDL in this study is more significant with R^2 of 0.8847 in the training phase and 0.8532 in the test phase, depicting a remarkable correlation between recorded and simulated values.
- When the performance index (PI), scatter index (SI) are taken into consideration, the HFDL model has a smaller value, leading to it surpassing LSTM, followed by FedGRU, T-GCN, DCRNN, and therefore is designated as the suggested model.
- In terms of statistical test with T_{stat} and U_{95} , the HFDL scenario explicitly has the lowest amounts of uncertainty when compared to other models, proving its superior generalization ability.
- Additionally, the results can be found that the model HFDL created based on federated learning can effectively improve the training efficiency, and its prediction accuracy can be comparable or even better than some centralized models in this study.

- The visualization results show that HF DL model can effectively explore the spatial–temporal dependencies of flight delays, providing guidance for formulating air traffic control strategies.

This study presents the capability of federated learning in flight delay prediction, which can conquer data sharing and preserve data privacy and security and has favorable prediction performance compared to conventional centralized learning methods. Various statistical evaluation metrics (R^2 , $RMSE$, MAE , VAF , PI , SI , T_{stat} , U_{95}) also confirm this. Yet our study is not without limitations. First, we do not include outlier identification as part of the data cleaning process. However, this issue deserves to be addressed by further research because raw flight delay log data are very noisy. In addition, flight delays may be affected by a wide range of external factors, such as weather conditions, air traffic control, etc., which have not yet been considered in the model. In future work, we are interested in investigating the use of the proposed model in scenarios where flight delays are dynamically changing. The adaptability of such models to various changing traffic conditions (e.g., air traffic control, emergencies such as epidemics, etc.), environments, and other factors could be investigated. We will further explore in depth various aspects of the field of flight delay prediction, including model optimization, consideration of incorporating spatial attention to enhance model performance, uncertainty modeling, and cross-domain collaboration. Additionally, when the flight datasets of more countries/regions are available, the model generalizability across different scenarios and geographical regions could be examined, and meanwhile the similarities and differences between these countries/regions could also be systematically compared. These studies will help to enhance the accuracy and reliability of flight delay prediction and provide better decision support and services to airlines, airports, and travelers.

Regarding the broader impact statement, although the proposed hybrid federated deep learning model (HF DL) demonstrates excellent performance in predicting network-wide flight delays, which provides airlines with more accurate information in flight scheduling and resource allocation, which helps to improve efficiency and reduce unnecessary impact on the environment. By reducing unnecessary flight waiting and ground handling time, airlines can save on fuel consumption, thereby reducing carbon emissions. There are also potential negative implications that must be considered. If model predictions are mistreated or abused, they could lead to irrational flight scheduling or passenger service decisions, which in turn could increase the carbon footprint of the aviation industry. In addition, the computational load on the central servers may increase as the models are widely used, leading to additional energy consumption and carbon emissions.

CRediT authorship contribution statement

Xiuyu Shen: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Jingxu Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization. **Ran Yan:** Writing – review & editing, Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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